

Project Testing

DATE	29 September 2026
TEAM ID	SWTID-2026-1812
PROJECT NAME	Comic craft Ai comic story creator using gemini model
MAXIMUM MARK	5 Mark

Performance Description

Performance testing is conducted to evaluate the responsiveness, stability, and efficiency of the ComicCraft – AI Comic Story Creator application. The testing focuses on the time taken by the application to load pages, generate story content, create comic images, display the preview, and generate the final PDF.

Testing Overview

Testing is an important phase of the ComicCraft – AI Comic Story Creator project. It is performed to verify that the application works correctly, meets the specified requirements, and provides the expected output to the user.

The ComicCraft application was tested by checking its main features, including user input, story generation, dialogue generation, image generation, comic preview, and PDF export.

Testing Objectives

- To verify that all major features work as expected.
- To identify and correct errors in the application.
- To check the accuracy and consistency of generated outputs.

- To verify that the application responds correctly to user inputs.
- To check the performance and stability of the application.
- To ensure that the final comic can be previewed and exported as a PDF.

Testing Types

S.N	Testing Type	Purpose
o		
1	Functional Testing	To verify that each feature performs its intended function.
2	Performance Testing	To evaluate application response and processing performance.
3	Integration Testing	To verify communication between the frontend, backend, Gemini AI, and image generation services.
4	Usability Testing	To check whether the application is simple and convenient for users.
5	Output Testing	To verify that the generated comic content and PDF are displayed correctly.

Testing Tools and Technologies

Tool / Technology	Purpose
FastAPI	Backend application testing

Google Gemini API	Testing AI-based story and dialogue generation
HTML / CSS / Jinja2	Testing user interface and comic preview
Python	Testing application logic and processing
Browser	Testing application pages and user interaction
PDF Export	Testing final comic document generation

Testing Result

The major functions of the ComicCraft application were tested successfully. The testing process helped verify that the application can accept user input, generate comic content using AI, display the generated panels, and export the final comic as a PDF.

Performance Parameters

S.No	Performance Parameter	Description
1	Page Loading	Checks whether the ComicCraft application pages load successfully and smoothly.
2	Story Generation	Checks the response of the application while generating the story outline and dialogue using Gemini AI.
3	Image Generation	Checks the performance of generating images for the comic panels.
4	Comic Preview	Checks whether the generated story, dialogue, and images are displayed correctly in the preview page.
5	PDF Generation	Checks whether the completed comic can be converted into a downloadable PDF successfully.

6	Server Response	Checks whether the FastAPI backend responds correctly to user requests.
7	Application Stability	Checks whether the application continues to function correctly during repeated generation requests.

Observation and Analysis

The ComicCraft application was tested under normal operating conditions. The main functions of the application, including story generation, dialogue generation, image generation, comic preview, and PDF generation, were tested.

The application successfully processed the required operations. The FastAPI server responded to user requests, and the generated content was displayed in the comic preview. The time required for AI-based operations may vary depending on network conditions and the response time of external AI services.

Performance Test Result

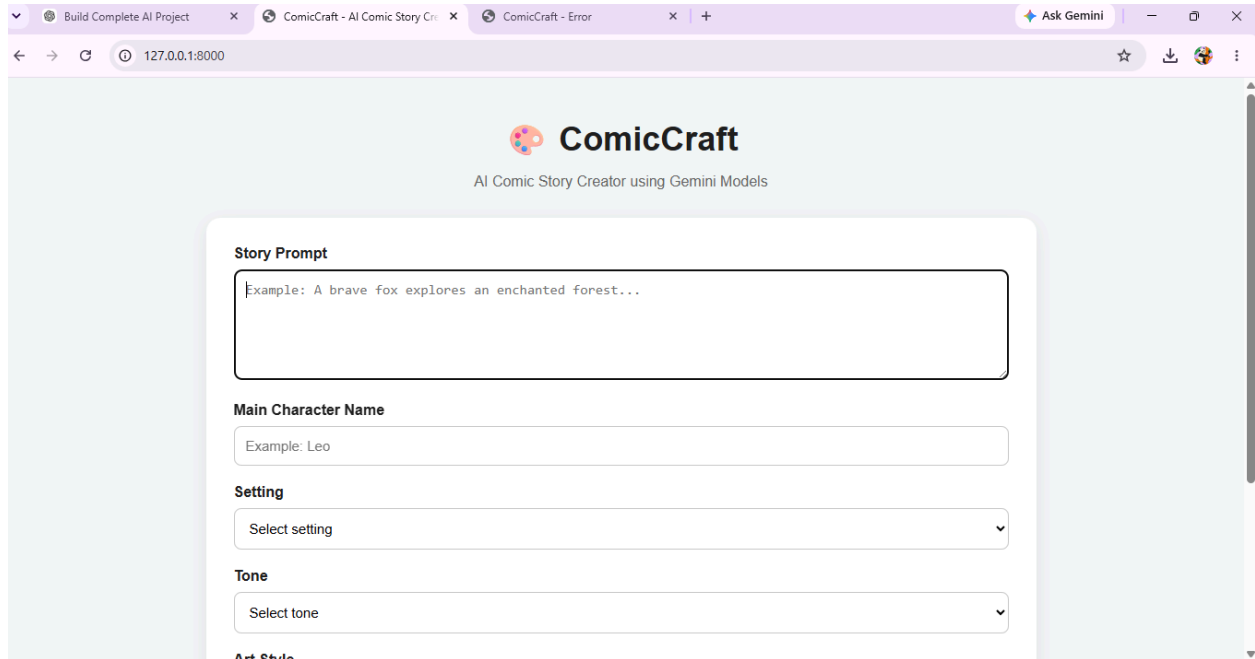
Test ID	Test Activity	Observation	Result
PT-01	Open Application	Home page loaded successfully.	Pass
PT-02	Generate Story	Story and dialogue were generated successfully.	Pass
PT-03	Generate Images	Comic panel images were processed and displayed.	Pass
PT-04	Preview Comic	Generated comic content was displayed correctly.	Pass
PT-05	Generate PDF	PDF was created successfully.	Pass
PT-06	Server Response	Backend responded to requests correctly.	Pass

Screenshot Analysis

The screenshots captured during testing provide evidence of the application's performance at different stages. The screenshots include the application home page, generated story output, comic preview, and PDF generation result.

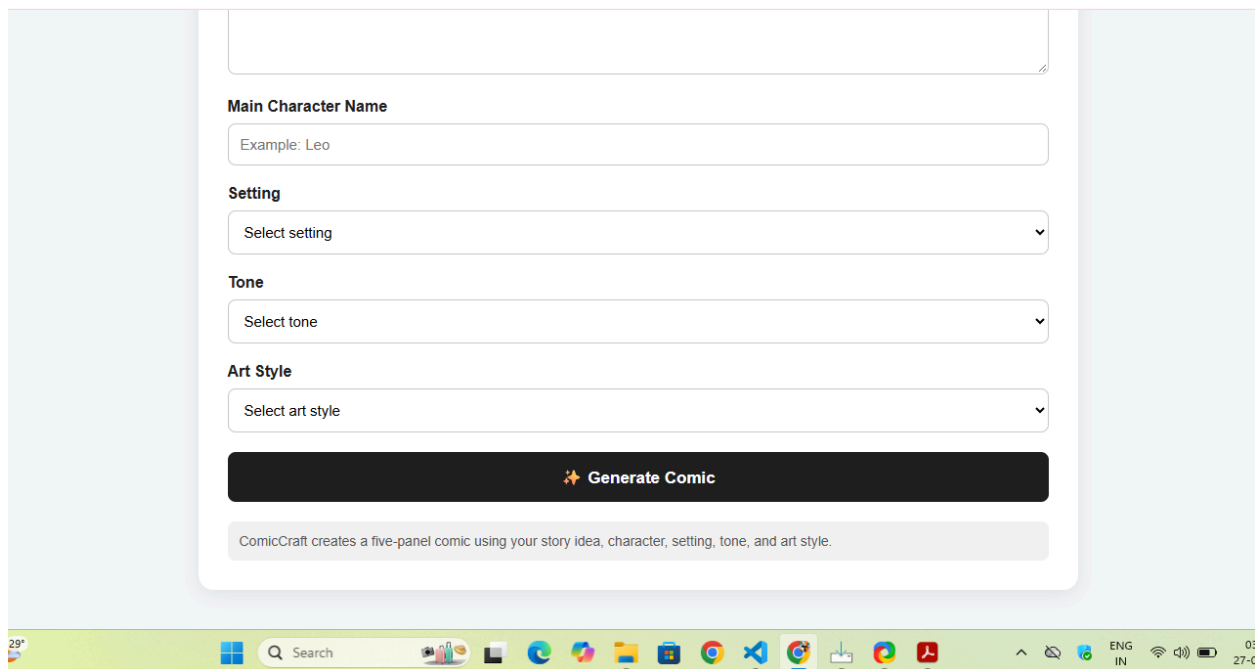
Screenshot 1 – Application Home Page

Shows that the ComicCraft application loads successfully and provides the required input fields.



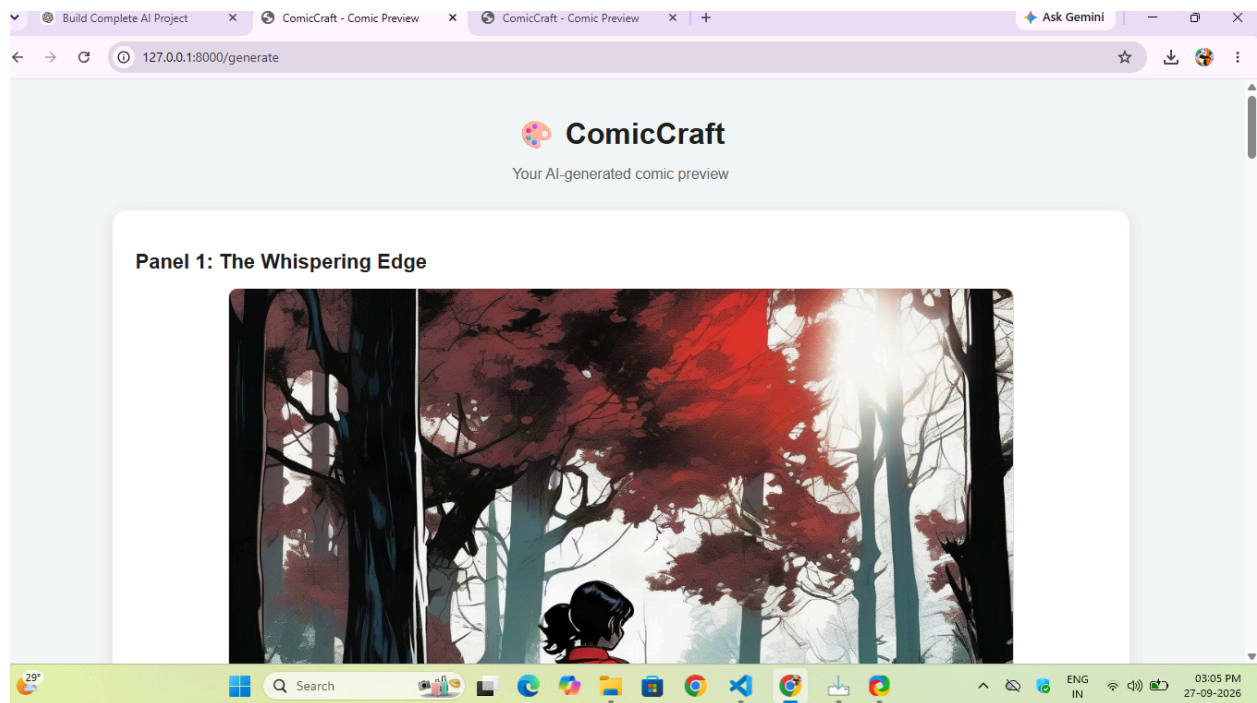
Screenshot 2 – Generated Story Output

Shows the story outline and dialogue generated from the user's prompt.



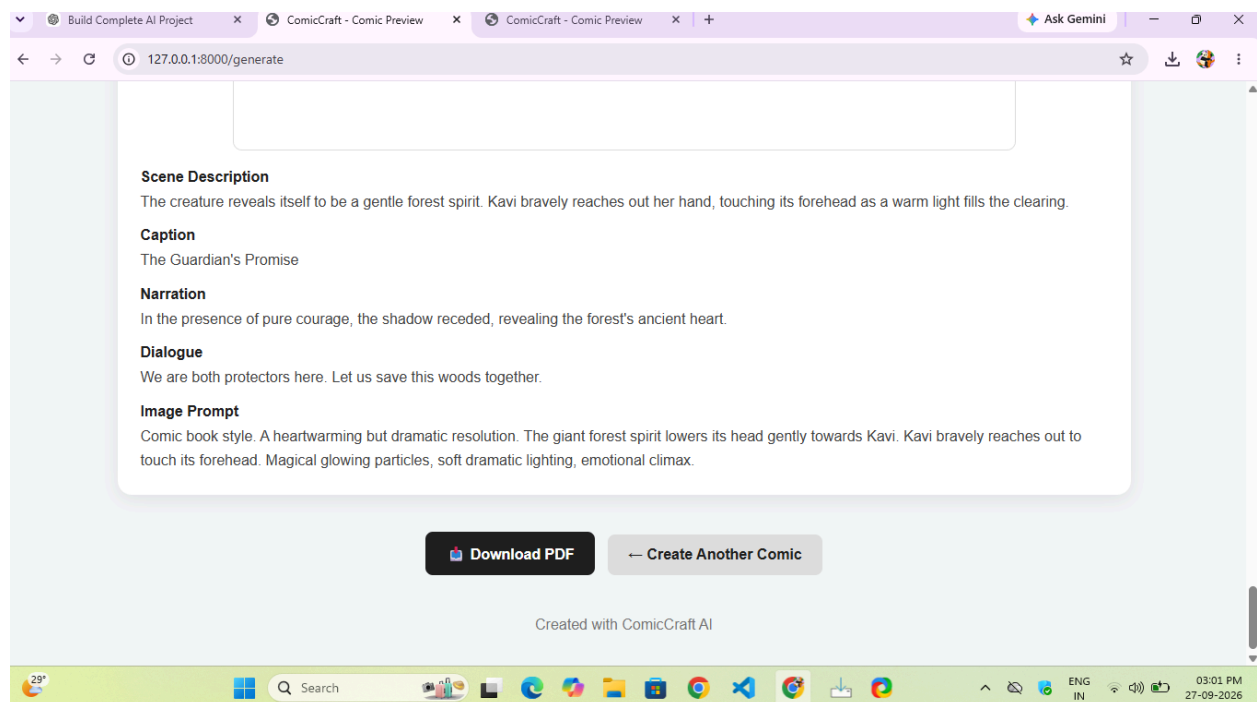
Screenshot 3 – Comic Preview

Shows the generated comic panels, narration, and dialogue in the preview page.



Screenshot 4 – PDF Generation

Shows the successful generation/download of the final comic PDF.



Conclusion

The performance testing confirms that the ComicCraft application performs its major functions successfully under normal testing conditions. The application provides a functional workflow from story input to AI-generated comic preview and PDF export.